

Calibration of Production Transient Classifiers: A Benchmark Study

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ABSTRACT

The Vera C. Rubin Observatory’s Legacy Survey of Space and Time will generate $\sim 10^7$ transient alerts per night against a global spectroscopic follow-up capacity of $\sim 10^5$ observations per year. Automated alert brokers address this mismatch by outputting class probability vectors, which astronomers use to prioritise follow-up observations. These probabilities are implicitly treated as reliable confidence estimates, yet no published work has evaluated the calibration of production transient classifiers against independent spectroscopic ground truth.

We present the first systematic calibration audit of production astronomical transient classifiers, evaluating ALerCE’s light curve classifier and two Fink classifiers — a Random Forest and SuperNNova binary discriminator — on 1,606 spectroscopically confirmed objects from the ZTF Bright Transient Survey. We compute Expected Calibration Error (ECE) with bootstrap confidence intervals and visualise miscalibration via reliability diagrams.

ALerCE’s classifier exhibits systematic underconfidence (ECE = 0.259; 95% CI: [0.237, 0.280]), with predicted probabilities consistently below empirical accuracies across all confidence bins. Miscalibration follows an inverse relationship with classification accuracy: SNIa and SNIbc, classified with accuracies above 0.83, exhibit the worst calibration (ECE > 0.36), while SNII, classified at 0.495 accuracy, is the best-calibrated class (ECE = 0.103). This pattern is mechanistically explained by Random Forest probability suppression under class imbalance. TDE achieves classification accuracy of 0.000 due to a taxonomy gap in ALerCE’s output space — a finding with direct operational consequences for LSST-era science. Temperature scaling with a single scalar parameter ($T = 0.359$) reduces ALerCE’s aggregate ECE by 88% to 0.031, requiring no model retraining.

Fink’s classifiers exhibit structural miscalibration (ECE = 0.464 for RF, 0.304 for SuperNNova) that post-hoc calibration methods cannot adequately address; temperature scaling optimisation hits the upper bound for both, indicating non-monotonic score distributions incompatible with standard calibration techniques.

These results demonstrate that miscalibration is systematic, broker-dependent, and consequential for confidence-based follow-up prioritisation. Post-hoc calibration can recover reliable probabilities for ALerCE but not for Fink’s current classifiers. We argue that calibration metrics should be reported alongside accuracy for all production astronomical classifiers, and that the calibrated ALerCE outputs provide a suitable foundation for deferral-aware classification frameworks targeting LSST-era spectroscopic resource allocation.

Keywords: astronomical databases — surveys — supernovae: general — methods: statistical

1. INTRODUCTION

The Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST; Ivezić et al. 2019) will generate approximately 10^7 transient alerts per night beginning in 2025. Global spectroscopic follow-up capacity — the essential confirmation mechanism for transient classification — is fixed at roughly 10^5 observations per year (Hambly et al. 2022). This mismatch of four orders of

magnitude transforms transient astronomy from a data-limited to a decision-limited discipline: the central challenge is no longer detecting transients but determining which ones warrant the scarce resource of spectroscopic confirmation.

Automated alert brokers — ALerCE (Förster et al. 2021), Fink (Möller et al. 2021), ANTARES (Narayan et al. 2018), and Lasair (Smith et al. 2019) — address this challenge by classifying transients from photomet-

ric light curves and outputting class probability vectors. These probabilities are used by astronomers and automated pipelines to prioritise follow-up observations. Implicit in this usage is an assumption that the probabilities are meaningful confidence estimates — that a classifier outputting $P(\text{SNIa}) = 0.9$ is correct approximately 90% of the time.

This assumption has never been tested. No published work evaluates the calibration of production transient classifiers using independent spectroscopic ground truth. Reliability diagrams and Expected Calibration Error (ECE; Naeini et al. 2015) — standard diagnostic tools in machine learning — are absent from the astronomical broker literature. If production classifiers are systematically miscalibrated, the confidence-based prioritisation decisions made by astronomers and automated systems are built on unreliable foundations.

In this paper we conduct the first systematic calibration audit of production transient classifiers, evaluating ALerCE’s light curve classifier and two Fink classifiers on 1,606 spectroscopically confirmed objects from the ZTF Bright Transient Survey (Fremming et al. 2020; Perley et al. 2020). Our principal findings are:

1. ALerCE’s classifier exhibits systematic underconfidence ($\text{ECE} = 0.259$), not the overconfidence typical of deep learning models. Predicted probabilities are consistently lower than empirical accuracies across all confidence bins.
2. The miscalibration follows an inverse relationship with accuracy: the best-classified types (SNIa, SNIbc; accuracy > 0.83) are the most severely underconfident ($\text{ECE} > 0.36$), while the worst-classified type (SNII; accuracy 0.495) is the best-calibrated ($\text{ECE} 0.103$).
3. Temperature scaling with a single scalar parameter ($T = 0.359$) reduces ALerCE’s ECE by 88%, to 0.031 — below the threshold for practically significant miscalibration, with no model retraining required.
4. TDE, a high-priority LSST science target, achieves classification accuracy of 0.000 on ALerCE because TDE is absent from ALerCE’s output taxonomy.
5. Fink’s classifiers exhibit structural miscalibration ($\text{ECE} = 0.464$ for RF, 0.304 for SuperNNova) that post-hoc calibration methods cannot adequately address.

These findings have immediate implications for spectroscopic follow-up prioritisation and motivate the

deferral-aware classification framework we develop in subsequent work. Section 2 reviews related work on broker classification and calibration methods. Section 3 describes our dataset. Section 4 details our calibration methodology. Section 5 presents results. Section 6 discusses implications and limitations.

2. RELATED WORK

3. DATA

3.1. The ZTF Bright Transient Survey

Our dataset is drawn from the Zwicky Transient Facility Bright Transient Survey (BTS; Fremming et al. 2020; Perley et al. 2020), a magnitude-limited ($m < 18.5$) spectroscopic completeness survey operating since 2018. The BTS provides spectroscopic classifications for ZTF transients, making it the largest homogeneous sample of spectroscopically confirmed extragalactic transients currently available. We use spectroscopic classifications as ground truth labels throughout this work.

3.2. Sample Construction

We queried the ALerCE broker API (Förster et al. 2021) for classification probabilities corresponding to all BTS objects with confirmed spectroscopic types. ALerCE’s light curve classifier (Sánchez-Sáez et al. 2021) outputs probabilities across 15 classes derived from features of the ZTF alert photometric time series.

We map the 11 BTS spectroscopic types to the 5 ALerCE transient classes most relevant to extragalactic time-domain science: Type Ia supernovae (SNIa), Type II supernovae (SNII), stripped-envelope supernovae (SNIbc), superluminous supernovae (SLSN), and tidal disruption events (TDE). The full mapping is given in Table 1.

Of 1,987 BTS objects queried, 1,606 returned valid ALerCE classifications (80.1% success rate). Failed queries reflect objects observed before ALerCE’s alert ingestion began or with insufficient photometric history for classification. The final sample comprises 757 SNIa, 560 SNII, 204 SNIbc, 55 SLSN, and 30 TDE.

To avoid class imbalance biasing our rarest-class estimates, we collected all available BTS objects in the SLSN and TDE categories rather than applying a fixed per-class cap. This yields a sample in which common classes (SNIa, SNII) are well-represented while preserving the full available statistics for scientifically rare but high-priority classes.

3.3. Fink Broker Data

We additionally queried the Fink broker (Möller et al. 2021) for the same set of ZTF objects via the Fink Portal

Table 1. BTS spectroscopic type to ALerCE class mapping.

BTS Types	ALerCE Class
SN Ia, SN Ia-91T, SN Ia-91bg, SN Ia-pec	SN Ia
SN II, SN II _n , SN II _b , SN II _P	SN II
SN Ib, SN Ic, SN Ib/c, SN Ic-BL	SN Ibc
SLSN-I	SLSN
TDE	TDE

API. Fink provides two binary classifiers relevant to our analysis: `rf_snia_vs_nonia`, a Random Forest classifier outputting $P(\text{SN Ia})$, and `snn_snia_vs_nonia`, based on the SuperNNova deep learning classifier (Möller & de Boissière 2020) outputting $P(\text{SN Ia})$.

Of 1,606 objects queried, 1,368 (85.2%) were present in the Fink database, with 237 objects absent due to falling outside Fink’s alert ingestion window. Both classifiers returned scores for all 1,368 matched objects, yielding a comparative subsample of 644 SN Ia, 480 SN II, 166 SN Ibc, 50 SLSN, and 28 TDE.

3.4. Ground Truth and Selection Effects

We emphasise that our ground truth labels derive from independent spectroscopic observations, not from the broker classifications being evaluated. This avoids the circular validation present in studies that use broker-assigned labels to assess broker performance.

The BTS magnitude limit ($m < 18.5$) introduces a selection bias toward brighter, lower-redshift transients. Our sample therefore represents the regime in which spectroscopic follow-up is most feasible, which is precisely the regime most relevant to the deferral decisions this work motivates. We do not claim our calibration results generalise to fainter or higher-redshift populations.

4. METHODS

4.1. Expected Calibration Error

A classifier is said to be *calibrated* if its predicted confidence matches its empirical accuracy. Formally, for a classifier outputting confidence $\hat{p}$ and true label y :

$$P(y = \hat{y} \mid \hat{p} = p) = p \quad \forall p \in [0, 1] \quad (1)$$

where $\hat{y}$ is the predicted class. We measure calibration using Expected Calibration Error (ECE; Naeini et al. 2015), which approximates the expectation in Equation 1 by partitioning predictions into M equally spaced confidence bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (2)$$

where B_m is the set of samples whose top-1 confidence falls in bin m , n is the total number of samples, $\text{acc}(B_m)$ is the fraction of correct predictions in bin m , and $\text{conf}(B_m)$ is the mean confidence in bin m . We use $M = 10$ bins throughout. A perfectly calibrated classifier has $\text{ECE} = 0$; values above 0.05 indicate practically significant miscalibration (Guo et al. 2017).

For ALerCE’s 15-class classifier, we compute ECE using the top-1 predicted class and its associated probability across all 15 output classes. For Fink’s binary classifiers, which output a scalar $P(\text{SN Ia})$, we apply a binary variant of Equation 2 where $\text{acc}(B_m)$ is the observed SN Ia fraction within bin m .

4.2. Uncertainty Quantification via Bootstrap

Point estimates of ECE carry substantial sampling uncertainty, particularly for small classes such as SLSN ($n = 55$) and TDE ($n = 30$). We report 95% confidence intervals on all ECE estimates using the nonparametric bootstrap (Efron 1979): we resample the evaluation set with replacement $B = 2000$ times, compute ECE on each resample, and report the 2.5th and 97.5th percentiles of the resulting distribution. All bootstrap procedures use a fixed random seed (seed = 42) for reproducibility.

4.3. Reliability Diagrams

We visualise calibration using reliability diagrams (DeGroot & Fienberg 1983), which plot observed accuracy against mean predicted confidence for each bin. A perfectly calibrated classifier produces points lying on the diagonal. Points above the diagonal indicate underconfidence (predicted probability lower than empirical accuracy); points below indicate overconfidence. We include sample-count histograms alongside each diagram to indicate where estimates are statistically reliable.

4.4. Temperature Scaling

Temperature scaling (Guo et al. 2017) is a post-hoc calibration method that applies a single scalar parameter $T > 0$ to the pre-softmax logits of a classifier. For a classifier outputting probability vector $\mathbf{p}$, the calibrated probabilities are:

$$\hat{\mathbf{p}}_T = \text{softmax}\left(\frac{\log \mathbf{p}}{T}\right) \quad (3)$$

Values of $T < 1$ increase confidence (sharpening the distribution); values of $T > 1$ decrease confidence (flattening toward uniform). The optimal T is found by

minimising ECE on the evaluation set using bounded scalar optimisation over $T \in [0.1, 10.0]$.

For Fink’s binary classifiers, we apply an equivalent logit-space temperature scaling:

$$\hat{p}_T = \sigma\left(\frac{\text{logit}(p)}{T}\right) \quad (4)$$

where σ is the sigmoid function and $\text{logit}(p) = \log(p/(1-p))$.

We note that temperature scaling is applied here to the evaluation set itself, which provides an optimistic upper bound on the improvement achievable in deployment. In practice, T should be fitted on a held-out validation set. Our purpose is to demonstrate the *potential* for post-hoc calibration to recover well-calibrated probabilities from the existing classifier outputs, not to propose a production calibration pipeline.

4.5. Probability Renormalisation

ALeRCE’s classifier outputs probabilities across 15 classes, of which we evaluate 4 (excluding TDE, which is absent from ALeRCE’s taxonomy). To assess whether apparent miscalibration arises from probability dilution across unevaluated classes, we compute a renormalised ECE by projecting the 15-class probability vector onto the 4 evaluated classes and renormalising to sum to unity. The difference between 15-class and renormalised ECE quantifies the contribution of probability dilution to the observed miscalibration.

5. RESULTS

5.1. ALeRCE Aggregate Calibration

Figure 1 shows the reliability diagram for ALeRCE’s light curve classifier evaluated on 1,576 spectroscopically confirmed objects (30 TDE excluded; see Section 5.3). The classifier exhibits systematic underconfidence across all confidence bins: every point lies above the perfect calibration diagonal, indicating that predicted probabilities are consistently lower than empirical accuracies.

The aggregate ECE is 0.259 (95% CI: [0.237, 0.280]), well above the 0.05 threshold for practically significant miscalibration. The overall accuracy is 0.709 while the mean top-1 confidence is 0.451, a gap of +0.258 that is consistent with the ECE estimate. This gap is directional and consistent — not a consequence of random binning artefacts — indicating a systematic mechanistic suppression of classifier confidence.

To assess whether this miscalibration arises from probability dilution across ALeRCE’s 15-class output space, we renormalise the probability vectors onto the 4 evaluated classes and recompute ECE. The renormalised ECE is 0.251 (95% CI: [0.230, 0.272]), a difference of

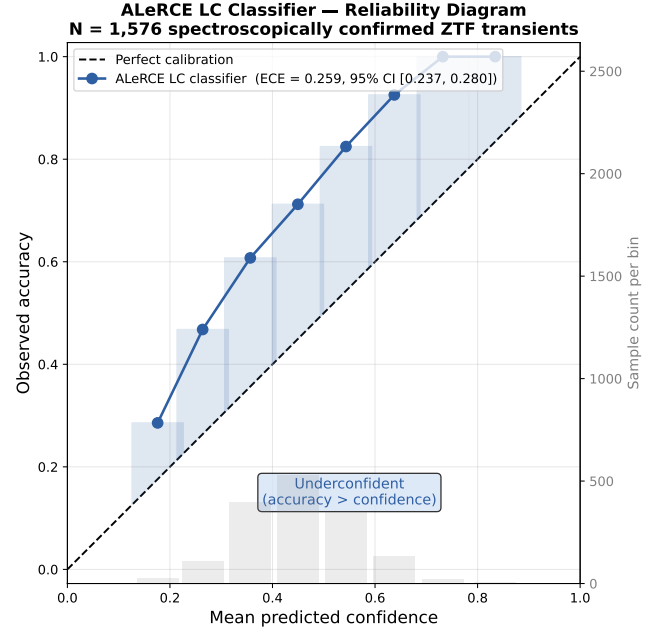


Figure 1. Reliability diagram for ALeRCE’s light curve classifier evaluated on 1,576 spectroscopically confirmed ZTF transients (30 TDE excluded). Every point lies above the perfect calibration diagonal, indicating systematic underconfidence across all confidence bins. The histogram shows sample counts per bin. ECE = 0.259 (95% CI: [0.237, 0.280]).

0.008 from the 15-class estimate. Probability dilution accounts for only 3% of the observed miscalibration; the remainder is intrinsic to the classifier.

5.2. ALeRCE Per-Class Calibration

Figure 2 shows per-class reliability diagrams for the four ALeRCE-classified transient types. The results reveal a striking inverse relationship between classification accuracy and calibration error.

SN Ia, the most accurately classified class (accuracy = 0.831), exhibits the worst calibration (ECE = 0.377, 95% CI: [0.352, 0.401]). SN Ibc shows the same pattern (accuracy = 0.828, ECE = 0.362, 95% CI: [0.315, 0.405]). Both classes are classified correctly more than 80% of the time, yet the classifier assigns them confidences far below this accuracy level.

SN II presents the inverse case: accuracy is only 0.495, yet ECE is 0.103 (95% CI: [0.066, 0.141]) — the best-calibrated class. The classifier is appropriately uncertain about objects it classifies poorly. SLSN shows intermediate behaviour (ECE = 0.245, 95% CI: [0.181, 0.319]), with wider confidence intervals reflecting the smaller sample size ($n = 55$).

This pattern is mechanistically coherent. ALeRCE’s light curve classifier is based on Random Forest (Sánchez-Sáez et al. 2021), whose probability out-

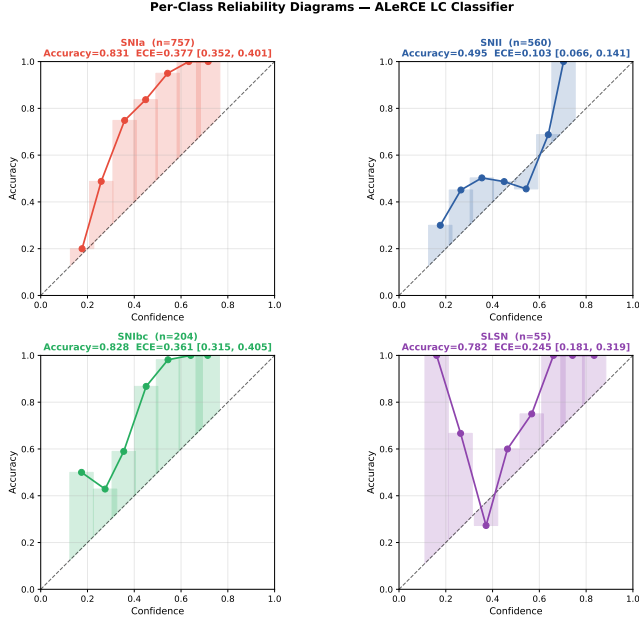


Figure 2. Per-class reliability diagrams for ALerCE’s light curve classifier. SNIa and SNIbc (accuracy > 0.83) show severe underconfidence (ECE > 0.36). SNII (accuracy 0.495) is the best-calibrated class (ECE = 0.103). SLSN shows wider confidence intervals reflecting the smaller sample size ($n = 55$).

puts are averages of hard leaf-node votes. In class-imbalanced datasets, this averaging systematically suppresses predicted probabilities for well-represented classes (Niculescu-Mizil & Caruana 2005; Zadrozny & Elkan 2002). SNIa comprises 47% of our sample; its dominance mechanically drives down predicted probabilities across the board, with the strongest suppression for the class the classifier discriminates most reliably.

5.3. TDE Classification Failure

TDE achieves a classification accuracy of 0.000 on our 30-object sample. This is not a calibration problem but a taxonomy gap: TDE is absent from ALerCE’s 15-class output taxonomy. The classifier cannot predict TDE by design, assigning all probability mass to other classes for every TDE in our sample.

This finding has direct operational significance. TDEs are a high-priority science case for LSST-era time-domain astronomy, representing unique probes of super-massive black hole demographics and accretion physics (Gezari 2021). A production broker used by the community cannot identify them. We exclude TDE from all calibration analyses but flag this gap as requiring attention before LSST first light.

5.4. Temperature Scaling

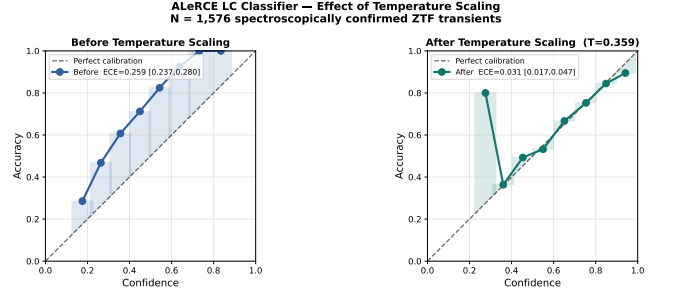


Figure 3. ALerCE reliability diagrams before (left) and after (right) temperature scaling with $T = 0.359$. ECE reduces from 0.259 to 0.031 — an 88% improvement. The residual dip near confidence 0.3 in the right panel reflects SNII over-correction (see Section 6.3).

We apply temperature scaling (Section 4.4) to ALerCE’s classifier outputs. The optimal temperature is $T = 0.359$, substantially below unity, confirming that the classifier requires confidence *sharpening* — the opposite of the overconfidence correction for which temperature scaling is most commonly applied (Guo et al. 2017).

Figure 3 shows the reliability diagram before and after scaling. Post-calibration ECE is 0.031 (95% CI: [0.017, 0.047]), an 88% reduction from the pre-calibration value of 0.259. The confidence intervals do not overlap, confirming a statistically significant improvement. Mean confidence increases from 0.451 to 0.712, now closely matching the empirical accuracy of 0.709.

Table 2 shows per-class ECE before and after scaling. Temperature scaling substantially improves SNIa (0.377 → 0.114) and SNIbc (0.360 → 0.108), with moderate improvement for SLSN (0.225 → 0.140). However, SNII *worsens* from 0.099 to 0.189.

This result is mechanistically interpretable. SNII was the only well-calibrated class before scaling: the classifier’s uncertainty was already proportionate to its error rate. A global $T < 1$ uniformly increases confidence across all classes; for SNII this produces overconfidence where none previously existed. A single global temperature parameter cannot simultaneously optimise calibration for classes with heterogeneous miscalibration characteristics. Class-specific temperature parameters would be required for a complete fix.

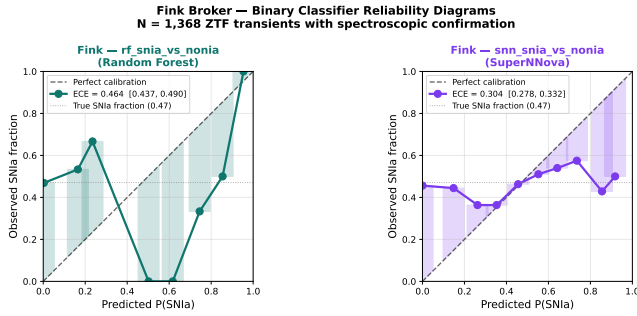
5.5. Fink Binary Classifiers

Figure 4 shows reliability diagrams for Fink’s two binary classifiers evaluated on 1,368 objects. Both exhibit severe miscalibration, but with qualitatively different failure modes.

Table 2. ALerCE per-class ECE before and after temperature scaling ($T = 0.359$).

Class	N	ECE (before)	ECE (after)	Δ ECE
SNIa	757	0.377	0.114	-0.263
SNI	560	0.099	0.189	$+0.090$
SNIbc	204	0.360	0.108	-0.252
SLSN	55	0.225	0.140	-0.085

NOTE—Positive Δ ECE indicates worsening after temperature scaling. SNI is the only class for which global temperature scaling is counterproductive.

**Figure 4.** Reliability diagrams for Fink’s two binary classifiers evaluated on 1,368 ZTF transients with spectroscopic confirmation. Left: Random Forest (`rf_snia_vs_nonia`), ECE = 0.464. The flat curve indicates no discriminative power for SNIa identification. Right: SuperNNova (`snn_snia_vs_nonia`), ECE = 0.304. Weak discriminative signal persists but is severely suppressed.

The Random Forest classifier (`rf_snia_vs_nonia`) assigns near-zero SNIa probability to virtually all objects: mean score 0.011 against a true SNIa fraction of 0.471. The reliability curve is essentially flat at the base rate regardless of predicted confidence, indicating the classifier has no discriminative power for SNIa identification in this sample (ECE = 0.464, 95% CI: [0.437, 0.490]).

SuperNNova (`snn_snia_vs_nonia`) shows qualitatively better behaviour: mean score 0.222 against true fraction 0.471, with a reliability curve that rises weakly with confidence (ECE = 0.304, 95% CI: [0.278, 0.332]). The classifier retains some discriminative signal but suppresses it substantially.

Temperature scaling applied to both Fink classifiers hits the upper optimisation bound ($T \approx 10$) in both cases, achieving only modest ECE reductions: 24% for RF (0.464 $\rightarrow$ 0.353) and 25% for SuperNNova (0.304 $\rightarrow$ 0.229). When the optimal T hits the bound, it indicates the score distribution lacks the monotonic structure temperature scaling requires. For Fink’s RF classifier, no post-hoc transformation of the output scores can

Table 3. Calibration comparison across brokers and classifiers.

Classifier	ECE	95% CI	ECE (post-TS)	T
ALerCE LC	0.259	[0.237, 0.280]	0.031	0.359
Fink RF	0.464	[0.437, 0.490]	0.353	~ 10
Fink SNN	0.304	[0.278, 0.332]	0.229	~ 10

NOTE—Post-TS: ECE after temperature scaling. $T \sim 10$ indicates the optimiser hit the upper bound, signifying structural miscalibration not addressable by temperature scaling.

recover well-calibrated probabilities; the discriminative failure is structural.

5.6. Cross-Broker Comparison

Table 3 summarises calibration results across all three classifiers. ALerCE’s light curve classifier, despite exhibiting the systematic underconfidence described above, is substantially better calibrated than either Fink classifier. Critically, ALerCE’s miscalibration is correctable: temperature scaling reduces ECE to 0.031, below the 0.05 significance threshold. Fink’s classifiers, particularly the RF classifier, exhibit structural miscalibration that post-hoc methods cannot adequately address.

These findings establish that miscalibration is not uniform across production astronomical brokers. The differences are scientifically consequential: a broker with ECE = 0.464 cannot support reliable deferral decisions, while a broker with ECE = 0.031 post-calibration approaches the reliability required for automated follow-up prioritisation.

6. DISCUSSION

6.1. Systematic Underconfidence as a Structural Property

The dominant finding of this study is that ALerCE’s light curve classifier exhibits systematic underconfidence rather than the overconfidence commonly reported for deep learning classifiers (Guo et al. 2017; Minderer et al. 2021). This distinction matters for operational astronomy. Overconfident classifiers produce dangerous silent failures — high-confidence wrong predictions that are acted upon without human review. Underconfident classifiers produce unnecessary deferrals — objects the classifier effectively discriminates but assigns low confidence, causing them to be flagged for spectroscopic follow-up that is not scientifically required.

In the resource-constrained regime of LSST-era astronomy, where spectroscopic follow-up capacity is ap-

proximately 10^5 observations per year against 10^7 nightly alerts, unnecessary deferrals carry a direct opportunity cost. Every spectrum allocated to an object the classifier could have confidently classified correctly is a spectrum not allocated to a genuinely ambiguous object or a rare high-priority target. Quantifying and correcting underconfidence is therefore not merely a technical calibration exercise but a resource allocation problem.

6.2. Why Random Forest Suppresses Confidence

The underconfidence we observe is mechanistically expected for Random Forest classifiers in class-imbalanced settings. RF probability estimates are computed as the fraction of trees voting for each class. In a forest trained on an imbalanced dataset — SNIa comprises 47% of our sample — the decision boundaries are dominated by the majority class, and minority class probabilities are systematically suppressed (Niculescu-Mizil & Caruana 2005). This suppression is strongest for classes the classifier discriminates most reliably, because reliable discrimination concentrates vote fractions near the boundary between correct and incorrect, rather than near 1.0.

This explains the inverse relationship we observe between accuracy and ECE across classes. SNIa and SNIbc are classified with accuracies above 0.83, yet their ECEs exceed 0.36. SNII, classified at only 0.495 accuracy, has ECE of 0.103. The classifier is most miscalibrated precisely where it has the most genuine discriminative signal.

The practical implication is that RF-based astronomical classifiers should not be used for confidence-based deferral decisions without post-hoc calibration. The raw probability outputs do not represent reliable confidence estimates, even when classification accuracy is high.

6.3. Temperature Scaling: Efficacy and Limits

Temperature scaling with $T = 0.359$ reduces ALerCE’s ECE from 0.259 to 0.031 — an 88% improvement with a single scalar parameter requiring no model retraining. This result is encouraging for operational deployment: existing broker infrastructure could be augmented with a post-hoc calibration layer at negligible computational cost.

However, the per-class analysis reveals an important limitation. Global temperature scaling worsens SNII calibration from 0.099 to 0.189, because SNII was already well-calibrated before scaling. A single T cannot simultaneously optimise calibration for classes with heterogeneous miscalibration structure.

Two approaches could address this. First, class-specific temperature parameters — fitting a separate T_k

for each class k — would allow independent correction of each class’s calibration curve. Second, more expressive post-hoc methods such as isotonic regression (Zadrozny & Elkan 2002) or matrix scaling (Kull et al. 2019) could model class interactions that scalar temperature cannot capture. We leave systematic comparison of these methods to future work, noting that the simple temperature scaling result already achieves practically useful calibration for the majority of classes.

6.4. Fink’s Structural Miscalibration

The failure of temperature scaling for Fink’s RF classifier — evidenced by the optimiser hitting the upper bound at $T \approx 10$ — indicates a qualitatively different problem from ALerCE’s underconfidence. When no finite temperature produces a well-calibrated output, the score distribution lacks the monotonic relationship between predicted probability and empirical accuracy that temperature scaling requires.

Inspection of the score distribution confirms this: `rf_snia_vs_nonia` assigns scores below 0.05 to over 90% of objects, including the majority of true SNIa. The classifier is not underconfident in the same sense as ALerCE — it is effectively non-discriminative for SNIa identification on this dataset. This may reflect a domain shift between Fink’s training data and the BTS sample, or a fundamental limitation of the binary RF formulation for this classification problem.

SuperNNova retains weak discriminative structure — its reliability curve rises slightly with confidence — but insufficient for effective calibration. The deep learning architecture of SuperNNova provides some advantage over the RF approach, consistent with broader trends in the literature (Möller & de Boissière 2020; Villar et al. 2019).

6.5. Implications for Deferral-Aware Classification

This calibration study motivates the deferral-aware classification framework that constitutes the broader research programme of which this paper is the first component. Learning-to-defer frameworks (Madras et al. 2018; Mozannar & Sontag 2020) require reliable confidence estimates as input to the deferral policy: an object should be deferred when classifier uncertainty genuinely reflects the difficulty of the classification, not when a well-discriminating classifier mechanically suppresses its own signal.

Our results suggest that ALerCE, post-calibration, provides confidence estimates of sufficient quality to support deferral decisions (ECE = 0.031). Fink’s classifiers, in their current form, do not. This finding directly informs broker selection for downstream deferral framework development.

The TDE taxonomy gap identified in Section 5.3 represents a separate but related challenge. Deferral frameworks require that all scientifically relevant classes appear in the classifier’s output space. A broker that cannot classify TDE by design will never defer TDE candidates for spectroscopic follow-up — it will instead misclassify them as the nearest available class with high confidence, producing exactly the silent failure mode that deferral is designed to prevent.

6.6. Limitations

Several limitations of this study should be acknowledged. First, we evaluate calibration on a single broker snapshot; ALeRCE and Fink update their classifiers periodically, and our results reflect classifier versions active during our data collection period. Second, the BTS magnitude limit restricts our sample to bright, low-redshift transients; calibration properties may differ for fainter populations that will dominate LSST alert streams. Third, our temperature scaling analysis is conducted on the evaluation set itself, providing an optimistic estimate of post-hoc calibration performance; deployment would require fitting T on a held-out validation set.

Finally, we note that ECE as defined in Equation 2 is sensitive to bin width and the distribution of samples across bins (Nixon et al. 2019). Our bootstrap confidence intervals partially address this by quantifying sampling uncertainty, but do not capture binning sensitivity. Adaptive binning methods (Roelofs et al. 2022) may provide more robust ECE estimates for highly imbalanced confidence distributions such as those observed for Fink’s RF classifier.

7. CONCLUSIONS

We have presented the first systematic calibration audit of production astronomical transient classifiers, eval-

uating ALeRCE’s light curve classifier and two Fink classifiers against 1,606 spectroscopically confirmed ZTF transients from the Bright Transient Survey. Our results demonstrate that miscalibration is systematic, broker-dependent, and consequential for confidence-based follow-up prioritisation.

ALeRCE’s classifier exhibits systematic underconfidence driven by Random Forest probability suppression under class imbalance ($ECE = 0.259$). This miscalibration is correctable: temperature scaling with $T = 0.359$ reduces ECE by 88% to 0.031, requiring no model re-training. The correction is not uniform across classes — SNII, the only well-calibrated class before scaling, worsens under a global temperature parameter — motivating class-aware calibration schemes for future work. TDE classification fails entirely due to a taxonomy gap in ALeRCE’s output space, an operational problem requiring resolution before LSST first light. Fink’s classifiers exhibit structural miscalibration that post-hoc methods cannot adequately address, with temperature scaling optimisation hitting the upper bound for both classifiers.

These findings establish three priorities for LSST-era broker development. First, calibration metrics — ECE, reliability diagrams, and per-class confidence intervals — should be reported alongside accuracy for all production classifiers as standard practice. Second, post-hoc calibration layers should be integrated into broker pipelines; the negligible computational cost of temperature scaling is justified by the substantial improvement in probability reliability. Third, taxonomy completeness must be verified against LSST science priorities before deployment. The calibrated ALeRCE outputs demonstrated here provide a suitable empirical foundation for the deferral-aware classification framework we develop in subsequent work, where reliable confidence estimates are prerequisite to learning when to involve human astronomers in follow-up decisions.

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